

RESEARCH ARTICLE

Time-Dependent Effects of Subsoil Chiselling on Soil Physical Resilience in a Semi-Arid Soybean-Wheat Cropping System

Mukhtar Ahmad^{1,2} | Pramila Aggarwal¹ | R. N. Garg¹ | Surajit Mondal^{1,3}  | Sunayan Saha^{1,4} | Debasish Roy^{1,5} | Ravender Singh¹ | Debashis Chakraborty^{1,6}

¹Division of Agricultural Physics, ICAR-Indian Agricultural Research Institute, New Delhi, India | ²India Meteorological Department, Meteorological Centre, Srinagar, UT of J&K, India | ³ICAR - Central Citrus Research Institute, Nagpur, India | ⁴ICAR-Central Potato Research Institute, Regional Station, Jalandhar, Punjab, India | ⁵Dr. Rajendra Prasad Central Agricultural University, Samastipur, Bihar, India | ⁶International Maize and Wheat Improvement Center-Bangladesh, Dhaka, Bangladesh

Correspondence: Debashis Chakraborty (debashisiari@gmail.com)

Received: 30 June 2025 | **Revised:** 16 October 2025 | **Accepted:** 4 December 2025

Keywords: early-season effects | field-saturated hydraulic conductivity | infiltration rate | penetration resistance | root growth | sub-surface compaction

ABSTRACT

Sub-surface soil compaction restricts root growth and water movement in intensively cultivated semi-arid croplands. Chiselling is commonly used to alleviate compaction, but its effects are often short-lived under variable rainfall conditions. We evaluated the time-variable effects of chiselling on soil hydro-mechanical properties (penetration resistance [PR], bulk density [BD], hydraulic function [steady-state infiltration, IR; field-saturated hydraulic conductivity, K_{fs}] and soil moisture dynamics), root traits (length, volume, surface area and diameter), yield and water use efficiency (WUE) in a soybean-wheat rotation on a sandy loam soil over 2 years. Four treatments were evaluated: residual chisel (RS; May 2011), repeated chisel (RC; May 2011 and May 2013), fresh chisel (FC; May 2013), and no chisel (NC). At early soybean 2013, the compact layer (15–30 cm) showed 20%–22% lesser PR in RS and RC than NC ($p = 0.019$); BD reductions mirrored this pattern. Despite greater PR, NC initially exhibited the highest IR (1.36 cm h^{-1} vs. RS 0.79 , RC 1.08 , FC 0.90 cm h^{-1}) and K_{fs} (1.10 – 1.19 cm h^{-1} , 31%–82% above chiselled plots at 15–30 cm), consistent with a dual-porosity mechanism in which a few continuous biopores govern near-saturated flow while the surrounding matrix remains strong. Root responses concentrated at 15–30 cm. RS/RC increased root length, volume and surface area and moderately enlarged mean diameter relative to NC/FC ($p < 0.05$). Yield gains and improved WUE were observed in soybean in year 1 under RS (21% and 10%) and RC (32% and 17%) versus NC, whereas wheat yield did not differ; WUE was modest. Rainfall timing explained the short-lived FC gains. The first 14 days after chiselling (20 May 2013) were dry, followed by two storm clusters within days 15–60 totaling ~330 mm, including ≥ 50 –100 mm single-day events. These high-load, antecedently wet events plausibly collapsed freshly created voids in FC, truncating hydraulic gains, while RS/RC retained some storage advantages. The findings highlight that the effectiveness of chiselling is strongly time- and rainfall-dependent, and their persistence depends on pore-network continuity. Results suggest site-specific, rainfall-informed tillage decisions to enhance soil structural resilience and sustainable productivity in semi-arid systems.

Summary

- Chiselling reduced sub-surface compaction, but benefits declined by year two.
- Residual chiselling improved root growth, water retention and first year soybean yield.
- Fresh chiselling failed after post-tillage storm clusters due to soil reconsolidation.
- Timing relative to high-intensity rainfall is critical; use a site-specific monitor-and-trigger plan.
- Infiltration and hydraulic conductivity were initially highest in NC (pore continuity) but attenuated over time.

1 | Introduction

Sub-surface compaction is a major soil degradation issue in intensively cultivated systems, limiting root growth, soil water redistribution and nutrient availability. It has been widely documented in the Indo-Gangetic Plains in India (Aggarwal et al. 2006), and globally (Hamza and Anderson 2005; Ohtomo and Andy Tan 2001; Russell et al. 2001).

With agricultural intensification, sub-surface compaction has developed gradually from regular mechanical operations over decades (Ehlers et al. 1983), yet it remains an often overlooked or poorly understood phenomenon (Keller and Or 2022; Ren et al. 2022). Compared with soil structural descriptors, the functional implications of sub-surface compaction are even less documented (Hu et al. 2021). Compaction below the plough layer (15–40 cm) restricts root proliferation and affects the movement of water and nutrients, key determinants of soil structural resilience.

The impact of compaction on water movement and redistribution is manifested directly through changes in hydraulic properties (Alaoui et al. 2011; Hansson et al. 2019) and indirectly through increased soil resistance to penetration, reduced aeration and restricted root water uptake (Horton et al. 1994). High soil penetration resistance (PR), typically exceeding 2 MPa, has been shown to significantly limit root growth and function (Silva et al. 2008; Lima et al. 2012). While soil bulk density (BD) is often used as a proxy, PR is considered a more dynamic indicator, reflecting interactions among soil moisture, texture and pore connectivity (To and Kay 2005; Dexter et al. 2007; Tavares Filho et al. 2012).

While many studies have highlighted the benefits of deep tillage for alleviating sub-soil compaction and enhancing crop performance, others raise concerns about structural disruption and re-compaction risks (Schneider et al. 2017). The positive effects of chiselling are often transient (Araujo et al. 2004; Busscher et al. 2002), and re-compaction due to settling of particles and loss of pore connectivity is likely, especially under high rainfall (Rao et al. 1998; Allen and Musick 2001). Such processes reduce field-saturated hydraulic conductivity (K_{fs}) and increase surface-runoff potential (Kincaid 2002).

Understanding the timing and persistence of chiselling effects, particularly the ‘residual’ versus ‘fresh’ effects is essential,

especially in semi-arid regions where rainfall variability can accelerate structural degradation. While short-term improvements in infiltration, root growth and crop yield can occur, responses are often inconsistent, depending on soil type, rainfall, cropping systems, extent of disturbance and the time since chiselling (Azooz et al. 1999; He et al. 2019; Ma et al. 2015). Studies indicate the structural benefits of chiselling are short-lived, typically 2–2.5 years before the compaction returns to previous level (Botta et al. 2006; de Moraes et al. 2016; Raper 2005). Consequently, the efficacy and persistence of deep tillage in alleviating compaction remain uncertain (Neuens and Reheul 2003; Piccoli et al. 2022).

Given the vulnerability of semi-arid soils to degradation and rainfall-driven reconsolidation, a time-variable assessment of chiselling is crucial. High-intensity rainfall shortly after chiselling has been reported to negate up to 60% of the structural benefits (Keim and Faiers 1996; Bengtson and Carter 2004), particularly if it precedes sowing. Although improvements in water retention immediately post-chiselling can encourage deeper rooting and crop water uptake, these benefits may rapidly diminish if soil re-compacts as the season progresses.

We hypothesised that residual and repeated chiselling treatments would sustain improvements in sub-surface soil structure and water dynamics longer than freshly applied chiselling under variable monsoonal rainfall, thereby supporting rooting and short-term crop performance.

2 | Materials and Methods

2.1 | Study Area, Climate and Soil

Field experiments were conducted in a soybean-wheat rotation at the ICAR-Indian Agricultural Research Institute, New Delhi (28°38'23"N and 77°09'27"E; 228.6 m a.s.l.). The climate is semi-arid, subtropical with hot dry summers and cold winters. Mean annual rainfall is 765 mm, of which ~75% is received during the south-west monsoon (July–September). During soybean (July–November), daily maximum temperatures were 25°C–39°C (2013) and 28°C–42°C (2014); during wheat (November–April) they were 12°C–41°C (2013–14) and 13°C–40°C (2014–15). Daily minimum temperatures were 13°C–28°C (2013) and 14°C–29°C (2014) in soybean and 10°C–22°C (2013–14) and –1°C to 25°C (2014–15) in wheat. Total rainfall during soybean was 1197 and 429 mm, and during wheat 164 and 316 mm in respective years.

The soil is a Typic Haplustept, sandy loam in texture at 0–15 cm and sandy clay loam below, alkaline (pH 7.8–8.1), low salinity (EC 0.71–0.83 dS m^{–1}), moderate in K_{fs} (1.01 cm h^{–1}) and medium in organic carbon (5.3 and 4.1 g kg^{–1} in 0–15 and 15–30 cm layers, respectively). Continuous cultivation over several decades has produced a compacted layer below ~10 cm (Mondal et al. 2019).

2.2 | Experimental Details

Four treatments were evaluated by time since chiselling in a randomised block design with six replicates (plot size 7.5 × 8.0 m):

(1) No chisel (NC)—no chiselling; (2) Residual chisel (RS)—chiselled once on 28 May 2011 (≈ 26 months before the first soybean crop); repeated chisel (RC)—chiselled on 28 May 2011 and again on 20 May 2013 (~ 15 days before sowing of soybean); and fresh chisel (FC)—chiselled on 20 May 2013. Chiselling was performed with a single tyne chisel plough to a depth of 30–40 cm in a criss-cross pattern using a 60 HP tractor, spaced at 50–120 cm intervals.

Soybean (var. Pusa 9814) was sown on 9 July 2013 and 4 July 2014 and harvested on 2 November 2013 and 29 October 2014. The crop was rainfed in 2013; in 2014, three irrigations (on 34, 74 and 96 days after sowing, DAS) were provided due to a prolonged dry period. Wheat (var. PBW-502) was sown on 26 November 2013 and 21 November 2014 and harvested on 23 April in the following years. The fertiliser doses were 20 kg N, 60 kg P_2O_5 and 40 kg K_2O ha^{-1} for soybean and 120, 60 and 40 kg N, P_2O_5 and K_2O ha^{-1} for wheat. Wheat received four irrigations at crown-root initiation, late tillering, booting and dough stages.

2.3 | Observations: Soil

Soil BD, PR, infiltration rate (IR) and K_{fs} were measured twice per season, at 45–60 DAS and at crop maturity (100–120 DAS). BD was determined by the core method (Veihmeyer and Hendrickson 1948) at 0–15, 15–30 and 30–45 cm. PR was measured with a Rimik cone penetrometer (Model CP20) at 15 mm intervals up to 45 cm. Data were logged in the field and transferred using CP20V2 software. Due to instrument malfunction, PR was not recorded at soybean harvest (2013) and at the initial stage of the following wheat crop (2013–14). The ponding method (Bouwer 1986) was employed to measure IR using a double-ring infiltrometer to steady state. K_{fs} was measured using a Guelph Permeameter (Model 2800 K1) at 0–15, 15–30 and 30–45 cm.

Surface (0–15 cm) soil water content was monitored gravimetrically and converted to volume wetness (SWC, % v/v) by multiplying with BD; for deeper layers (15–30, 30–45, 45–60, 60–90 and 90–120 cm) it was measured by a neutron probe (Model CPN-503 DR). SWC was monitored at 4–10 days intervals during crop growth.

2.4 | Observations: Roots

A root auger (7 cm diameter) was used to collect root samples at 80–85 DAS from 0 to 60 cm at 15 cm increments. The cores were quickly transferred to the laboratory, washed to separate from soil particles, twigs, etc., stored at 4°C, and subsequently scanned to obtain root length density (RLD), root volume density (RVD), root surface area density (RSAD) and root average diameter (RAD) for each layer.

2.5 | Observations: Grain Yield at Harvest

At maturity, a $2 \times 2 m^2$ area was harvested from each plot and sun-dried. Grains were separated from plants by thrashing and

winnowing by a mechanical thrasher and weighed. The grain yield was expressed as $kg ha^{-1}$. Crop water use was computed using the field-water balance method.

2.6 | Statistical Analysis

All analyses were performed in SAS (2006) and R (R Core Team 2024) for ANOVA as applicable to randomised block design. Treatments means were compared using Duncan's Multiple Range Test at 5% and 1% significance levels. We report means $\pm$ SE, exact p -values. Figures were prepared using MS Excel and R.

3 | Results

3.1 | Soil BD and PR

PR data confirmed the presence of a compact layer at 15–30 cm, with values exceeding 3000 kPa, ~ 1.2 times greater than in the ploughed layer (0–15 cm) (Table 1). During early soybean 2013, both RS and RC exhibited significantly lesser PR at 15–30 cm compared to NC. Specifically, PR values reduced to 2765 kPa in RC and 2701 kPa in RS, compared to 3449 kPa in NC ($p=0.019$). In contrast, the fresh chiselling (FC) was comparable to NC, indicating minimal impact on sub-surface compaction. PR data were unavailable at soybean harvest and during early wheat 2013–14 due to penetrometer malfunction, limiting interpretations during these periods.

Daily rainfall (May–August 2013), anchored on chiselling (20 May 2013), showed no rain in the first 14 days (0–14 days), followed by two storm clusters. In 15–30 days (3–18 June), rainfall totalled 144.6 mm with two very-heavy days (≥ 50 mm), including a 67.4 mm day on 17 June; the 3-day rolling total peaked at 133.8 mm (ending 18 June), and antecedent precipitation index, API ($\lambda=0.85$) rose to ~ 126 . In 31–60 days (19 June–18 July), another 186.8 mm fell with one very-heavy day (98.2 mm on 10 July) and a 3-day peak of 141.2 mm (ending 11 July). These storm clusters, rather than rainfall in the first fortnight, represent the principal loading events likely to accelerate reconsolidation of freshly loosened subsoil in FC (Figure S1; Table S1).

BD followed a similar pattern. In both the 15–30 and 30–45 cm layers, the BD was significantly lower in RS (1.61 and 1.63 $Mg m^{-3}$) and RC (1.60 and 1.64 $Mg m^{-3}$) ($p=0.031$) compared to that in NC (1.66 and 1.64 $Mg m^{-3}$) and FC (1.68 and 1.71 $Mg m^{-3}$). No significant differences in SWC were observed across treatments at these depths.

During wheat, no differences in BD were detected, although larger variations in SWC were recorded at 15–30 cm (16%–21%) and at 30–45 cm (19%–24%) than at surface ($\sim 19\%$) during early crop growth. Given the missing PR, no firm conclusions could be drawn. At harvest, neither PR nor BD differed significantly, suggesting diminishing chisel effects.

As shown in Table 2, the chiselling effect was not apparent in the second year, except early in wheat. PR was higher ($p=0.026$) in NC (3001 kPa) than RC (2552 kPa) at 15–30 cm, but was similar (2161 kPa, $p=0.78$) at 30–45 cm compared to RS (1860 kPa)

TABLE 1 | Penetration resistance (PR, kPa), bulk density (BD, Mg m⁻³), and moisture content (MC, % m⁻³) at three depths (0–15, 15–30 and 30–45 cm) under residual chisel (RS), repeated chisel (RC), fresh chisel (FC) and no chisel (NC) for soybean 2013 (Initial, Final) and wheat 2013–14 (Initial, Final).

Parameters	Depth (cm)								
	0-15			15-30			30-45		
	BD	PR	MC	BD	PR	MC	BD	PR	MC
Treatments									
Soybean 2013: Initial									
RS	1.54 (±0.02)	2453 (±236)	16.0 (±2.0)	1.61 (±0.02)	2701 (±84)	15.5 (±2.3)	1.63 (±0.02)	2215 (±86)	15.0 (±3.3)
RC	1.55 (±0.02)	2439 (±178)	16.3 (±2.1)	1.60 (±0.02)	2765 (±368)	16.0 (±2.7)	1.64 (±0.01)	2251 (±135)	15.7 (±1.5)
FC	1.55 (±0.03)	2911 (±155)	16.9 (±2.5)	1.68 (±0.01)	3278 (±183)	15.8 (±2.5)	1.71 (±0.03)	2879 (±187)	13.8 (±3.7)
NC	1.56 (±0.01)	2992 (±115)	16.5 (±2.3)	1.66 (±0.02)	3449 (±185)	14.6 (±3.0)	1.64 (±0.02)	3048 (±262)	13.7 (±3.0)
p	0.969	0.078	0.47	0.031	0.019	0.53	0.048	0.012	0.44
Soybean 2013: Final									
RS	1.65 (±0.01)	—	14.6 (±3.3)	1.72 (±0.01)	—	17.8 (±3.9)	1.72 (±0.01)	—	15.8 (±3.6)
RC	1.67 (±0.02)	—	14.0 (±1.5)	1.70 (±0.02)	—	17.7 (±2.3)	1.72 (±0.01)	—	14.8 (±1.9)
FC	1.64 (±0.02)	—	14.2 (±3.7)	1.77 (±0.001)	—	18.2 (±2.8)	1.78 (±0.01)	—	15.7 (±3.3)
NC	1.65 (±0.01)	—	14.6 (±3.1)	1.75 (±0.01)	—	16.0 (±1.6)	1.77 (±0.02)	—	14.4 (±1.6)
p	0.621	—	0.87	0.002	—	0.84	0.03	—	0.79
Wheat 2013-14: Initial									
RS	1.56 (±0.02)	—	20.3 (±2.7)	1.62 (±0.03)	—	20.5 (±2.5)	1.57 (±0.03)	—	23.1 (±1.1)
RC	1.57 (±0.02)	—	19.3 (±0.9)	1.62 (±0.04)	—	19.4 (±3.2)	1.58 (±0.04)	—	24.0 (±2.9)
FC	1.61 (±0.04)	—	18.8 (±1.1)	1.64 (±0.02)	—	16.0 (±1.8)	1.60 (±0.02)	—	20.1 (±3.4)
NC	1.52 (±0.03)	—	19.0 (±2.5)	1.59 (±0.01)	—	17.0 (±3.2)	1.66 (±0.03)	—	19.0 (±5.2)
p	0.233	—	0.08	0.714	—	0.3	0.278	—	0.21
Wheat 2013-14: Final									
RS	1.61 (±0.01)	2413 (±48)	13.0 (±2.2)	1.65 (±0.04)	2778 (±78)	14.4 (±3.0)	1.64 (±0.03)	2504 (±73)	11.4 (±3.0)
RC	1.62 (±0.02)	2363 (±113)	14.1 (±1.4)	1.65 (±0.01)	2750 (±137)	13.0 (±0.9)	1.58 (±0.02)	2350 (±87)	10.9 (±1.3)
FC	1.61 (±0.03)	2689 (±190)	12.7 (±0.6)	1.64 (±0.03)	2953 (±182)	11.4 (±1.6)	1.68 (±0.01)	2630 (±236)	10.3 (±1.3)
NC	1.60 (±0.03)	2776 (±25)	13.2 (±1.5)	1.59 (±0.04)	3164 (±101)	12.1 (±2.0)	1.65 (±0.03)	2579 (±236)	11.1 (±0.7)
p	0.88	0.134	0.882	0.84	0.188	0.84	0.761	0.435	0.86

Note: Values are means (±SE); the bottom row in each sub-panel reports treatment *p*-values for PR, BD and MC at each depth. 'Initial' denotes early-season and 'Final' denotes harvest-stage sampling; cells marked '—' indicate PR not recorded due to penetrometer malfunction.

and RC (1873 kPa). No significant BD differences were found across treatments.

3.2 | Steady-State Soil IR

Chiselling improved IR only at the initial stage of soybean in the first year. Unexpectedly, NC plots exhibited greater IR (1.36 cm h^{-1} ; $p=0.008$) compared to RS (0.79 cm h^{-1}), RC (1.08 cm h^{-1}) and FC (0.90 cm h^{-1}) plots (Table 3). Although counter-intuitive, greater IR in NC can co-occur with higher PR. Undisturbed soil likely contains a few, stable, vertically continuous biopores that connect the surface to subsoil. These 'preferential' flow paths dominate near-saturated infiltration even when the surrounding matrix is strong and resists penetration. In FC, chiselling created many cracks/voids but broke vertical continuity, so surface water encountered more dead ends and IR was not sustained despite lower PR. Differences were non-significant in subsequent crops or years, indicating attenuation of chisel effect. This rapid attenuation coincides with the mid-June storm cluster (max 1-day 67.4 mm; max 3-day 133.8 mm; API $\approx$ 126), consistent with collapse of newly formed macropores in FC under high, antecedent wet loading (Figure S1; Table S1).

3.3 | Field-Saturated Hydraulic Conductivity

Like IR, K_{fs} was greater in NC at 15–30 cm during the first soybean season (Figure 1). At early growth and harvest, NC recorded 1.04 ($p=0.011$) and 1.19 ($p=0.044$) cm h^{-1} , respectively, which were 55%–82% and 31%–44% greater than the chiselled plots. K_{fs} remained greater in NC (1.29 cm h^{-1}) in the succeeding wheat crop compared to RS (0.75 cm h^{-1}), RC (0.73 cm h^{-1}) and was similar to FC (0.92 cm h^{-1}). In the second year, K_{fs} converged across treatments (0.45 – 0.53 cm h^{-1}), indicating loss of chiselling benefits over time.

The K_{fs} pattern mirrors IR for the same reason. In NC, a small number of long, continuous macropores at 15–30 cm offers fast drainage routes. In FC, chiselling increased pore number and lowered PR in the matrix, but many pores were shorter or less connected and later collapsed after the June–July storms. Consequently, K_{fs} decreased even though PR was initially reduced.

3.4 | Profile Soil Water Content

Variation in SWC (%v/v) was most evident at 0–15 cm layer and minimal below 60 cm depth (Figure 2). In soybean, SWC was 19%–28% greater in RC and RS plots on 43 and 48 DAS, and 50%–70% greater on 57 and 72 DAS at 0–15 and 15–30 cm. At 30–45 cm, RC and RS exceeded NC on 30, 43, 52, 57 and 82 DAS, but were comparable to FC, except on 57 and 82 DAS. On average, NC had 15%–20% SWC in this layer, compared to 20%–24% in RS and RC. In deeper layers, treatment differences were marginal, except on 57 DAS at 45–60 cm, and on 82 DAS at 45–60 and 90–120 cm, where RC exceeded NC or FC.

In the wheat crop, differences were frequent. At 0–15 cm, significant differences occurred on 25 ($p=0.046$), 65 ($p=0.032$), 69 ($p<0.01$), 98 ($p<0.01$), 119 ($p=0.046$), 127 ($p=0.047$),

131 ($p<0.01$) and 135 ($p<0.01$) DAS. In most cases, RC had greater SWC, followed by RS. FC was similar to NC and differed from RC on 65, 69, 98 DAS and from RS only on 69 and 119 DAS. Across treatments, SWC decreased over time. On average, RS, RC, FC and NC recorded SWC of 13% (7–21), 14% (7–22), 12% (6–19) and 12% (6–20), respectively.

In the 15–30 cm layer, NC recorded lesser SWC compared to RS on 69, 131, and 135 DAS, and on 77 and 127 DAS compared to RC. FC had similar values to NC but was lesser than RS and/or RC on several dates. At 30–45 cm, six out of 19 observations showed significant treatment differences, with RS and RC being 2%–4% greater, while FC was comparable to or lesser than NC. In the 45–60 cm layer, chiselled plots exhibited greater SWC on 77, 83, 98 and 112 DAS (3%–4% greater in RC and 4%–5% greater in RS). At 60–90 cm, similar patterns were observed on 12 dates with FC close to NC. Finally, at 90–120 cm, differences were non-significant.

In the second cropping cycle, chiselling effects were evident only in a few days in the soybean crop. On those days, RC generally had greater SWC, followed by RS, while FC and NC were largely similar. In the subsequent wheat crop, chiselling effects were seldom evident, although RC recorded greater SWC on 63 and 94 DAS, while both RS and RC showed greater values on 110 DAS. This is consistent with pore geometry; RS/RC improves storage by loosening the matrix and adding pore volume, even if the network is less continuous for rapid through-flow. NC, by contrast, drains efficiently through a few continuous macropores (greater IR/ K_{fs}) but can retain less water on some dates. FC, freshly disturbed, was more vulnerable to storm-driven settling during June–July.

This is further illustrated in Figure 3, which shows the cumulative profile water (0–120 cm), expressed as cm of water, did not differ significantly among treatments, except for a marginally greater value in RC (20.6 cm) compared to NC (19.4 cm) with RS $\sim$ 20.0 cm and FC $\sim$ 19.8 cm.

3.5 | Root Growth Parameters

Effects of chiselling on roots were evident primarily in the first year, particularly in the 15–30 cm layer (Table 4). In soybean, RLD was greater in RS (0.17 cm cm^{-3}) and RC (0.18 cm cm^{-3}) than NC (0.13 cm cm^{-3}) and FC (0.11 cm cm^{-3}) ($p=0.01$). In wheat, RLD was also greater in RS (0.27 cm cm^{-3}) and RC (0.28 cm cm^{-3}) than NC (0.22 cm cm^{-3}) and FC (0.21 cm cm^{-3}). No treatment differences were observed in soybean for RVD and RSAD; however, in wheat, both parameters were significantly greater in RS and RC at 15–30 cm (RVD 1.63 and 1.60 cm cm^{-3} , $p=0.04$; RSAD 7.13 and $5.80 \text{ cm}^2 \text{ cm}^{-3}$, $p<0.01$). Chiselling also resulted in 7%–20% and 15%–17% increases in RSAD in the 0–15 and 15–30 cm layers, respectively, in wheat. In the second year of the rotation, the effects of chiselling were no longer evident.

3.6 | Yield and Water Use Efficiency

The yield of soybean was 32% and 20% higher in RC and RS, respectively, over NC (1351 kg ha^{-1}) in the first year. FC yielded the lowest among chiselled plots (1431 kg ha^{-1}) (Table 5). These gains

TABLE 2 | Penetration resistance (PR, kPa), bulk density (BD, Mg m⁻³), and moisture content (MC, % m⁻³) at three depths (0–15, 15–30 and 30–45 cm) under residual chisel (RC), repeated chisel (RS), fresh chisel (FC) and no chisel (NC) for soybean 2014 (Initial, Final) and wheat 2014–15 (Initial, Final).

Parameters	Depth (cm)								
	0–15			15–30			30–45		
	BD	PR	MC	BD	PR	MC	BD	PR	MC
Treatment									
				Soybean 2014: Initial					
RS	1.66 (±0.03)	2161 (±59)	15.0 (±2.9)	1.72 (±0.01)	2698 (±146)	13.9 (±1.6)	1.74 (±0.02)	1972 (±99)	11.1 (±1.2)
RC	1.70 (±0.03)	2173 (±74)	18.3 (±2.4)	1.75 (±0.03)	2559 (±101)	12.3 (±0.7)	1.76 (±0.02)	1830 (±51)	11.8 (±1.0)
FC	1.65 (±0.02)	2590 (±32)	17.1 (±1.6)	1.72 (±0.02)	2702 (±84)	16.2 (±0.4)	1.76 (±0.01)	2039 (±67)	12.0 (±3.2)
NC	1.63 (±0.02)	2641 (±91)	16.2 (±3.1)	1.74 (±0.01)	3124 (±121)	15.1 (±0.19)	1.77 (±0.01)	2054 (±29)	10.4 (±0.4)
p	0.255	0.291	0.76	0.349	0.071	0.03	0.56	0.052	0.51
				Soybean 2014: Final					
RS	1.67 (±0.01)	2140 (±97)	17.9 (±0.4)	1.71 (±0.01)	2898 (±126)	17.4 (±0.1)	1.76 (±0.01)	1972 (±83)	13.7 (±0.4)
RC	1.66 (±0.01)	2254 (±43)	18.1 (±0.4)	1.73 (±0.01)	2736 (±251)	17.4 (±0.8)	1.75 (±0.01)	1830 (±85)	13.7 (±1.2)
FC	1.65 (±0.01)	2307 (±65)	17.3 (±0.3)	1.71 (±0.01)	2883 (±74)	17.9 (±0.3)	1.74 (±0.02)	2039 (±76)	13.2 (±0.9)
NC	1.66 (±0.01)	2368 (±56)	17.3 (±0.7)	1.72 (±0.01)	3028 (±119)	16.7 (±0.3)	1.75 (±0.03)	2054 (±62)	13.2 (±1.02)
p	0.569	0.246	0.68	0.437	0.274	0.84	0.14	0.087	0.92
				Wheat 2014–15: Initial					
RS	1.58 (±0.03)	2486 (±72)	25.1 (±0.5)	1.66 (±0.04)	2713 (±108)	26.4 (±1.1)	1.71 (±0.01)	1860 (±64)	27.9 (±1.4)
RC	1.60 (±0.01)	2285 (±58)	26.3 (±1.2)	1.67 (±0.01)	2552 (±110)	28.0 (±0.9)	1.72 (±0.04)	1873 (±35)	27.2 (±2)
FC	1.62 (±0.02)	2257 (±82)	24.8 (±0.9)	1.69 (±0.01)	2772 (±50)	27.01 (±1.1)	1.72 (±0.01)	1946 (±14)	27.4 (±1)
NC	1.61 (±0.01)	2395 (±33)	24.0 (±0.5)	1.65 (±0.02)	3001 (±47)	26.0 (±0.9)	1.72 (±0.02)	2161 (±37)	26.0 (±2)
p	0.618	0.07	0.57	0.81	0.026	0.31	0.994	0.78	0.0002
				Wheat 2014–15: Final					
RS	1.65 (±0.01)	2259 (±110)	14.3 (±0.3)	1.70 (±0.02)	2365 (±231)	12.9 (±0.5)	1.73 (±0.01)	1867 (±101)	12.0 (±0.2)
RC	1.64 (±0.02)	2108 (±115)	13.8 (±0.2)	1.72 (±0.02)	2544 (±97)	12.1 (±0.5)	1.75 (±0.02)	1990 (±135)	12.0 (±0.7)
FC	1.65 (±0.01)	2027 (±34)	13.7 (±0.2)	1.71 (±0.01)	2871 (±33)	12.4 (±0.6)	1.75 (±0.01)	1861 (±130)	11.3 (±0.9)
NC	1.64 (±0.01)	2251 (±173)	13.4 (±0.3)	1.73 (±0.01)	2870 (±95)	13.3 (±0.3)	1.75 (±0.01)	2035 (±156)	10.5 (±0.5)
p	0.8	0.545	0.57	0.52	0.076	0.31	0.657	0.701	0.78

Note: Values are means ($\pm$ SE); the bottom row in each sub-panel reports treatment p -values for PR, BD and MC at each depth. ‘Initial’ denotes early-season and ‘Final’ denotes harvest-stage sampling.

TABLE 3 | Steady state infiltration rates (cm h^{-1}) under residual chisel (RS), repeated chisel (RC), fresh chisel (FC) and no chisel (NC) for soybean and wheat crops.

Treatment	Soybean 2013		Wheat 2013–14		Soybean 2014		Wheat 2014–15	
	Initial	Final	Initial	Final	Initial	Final	Initial	Final
RS	0.79 (± 0.09)	1.34 (± 0.13)	0.95 (± 0.10)	1.14 (± 0.09)	1.17 (± 0.05)	1.05 (± 0.10)	0.92 (± 0.04)	0.80 (± 0.04)
RC	1.08 (± 0.18)	1.15 (± 0.19)	0.88 (± 0.09)	0.88 (± 0.09)	1.06 (± 0.08)	1.01 (± 0.15)	0.83 (± 0.06)	0.86 (± 0.01)
NC	1.36 (± 0.05)	1.48 (± 0.06)	1.10 (± 0.04)	1.08 (± 0.05)	1.11 (± 0.07)	1.07 (± 0.07)	0.91 (± 0.08)	0.85 (± 0.03)
FC	0.90 (± 0.07)	1.32 (± 0.09)	1.04 (± 0.06)	1.04 (± 0.09)	1.10 (± 0.10)	1.03 (± 0.06)	0.89 (± 0.07)	0.86 (± 0.05)
<i>p</i>	0.008	0.225	0.228	0.47	0.496	0.33	0.632	0.769

Note: Values are mean ($\pm$ SE).

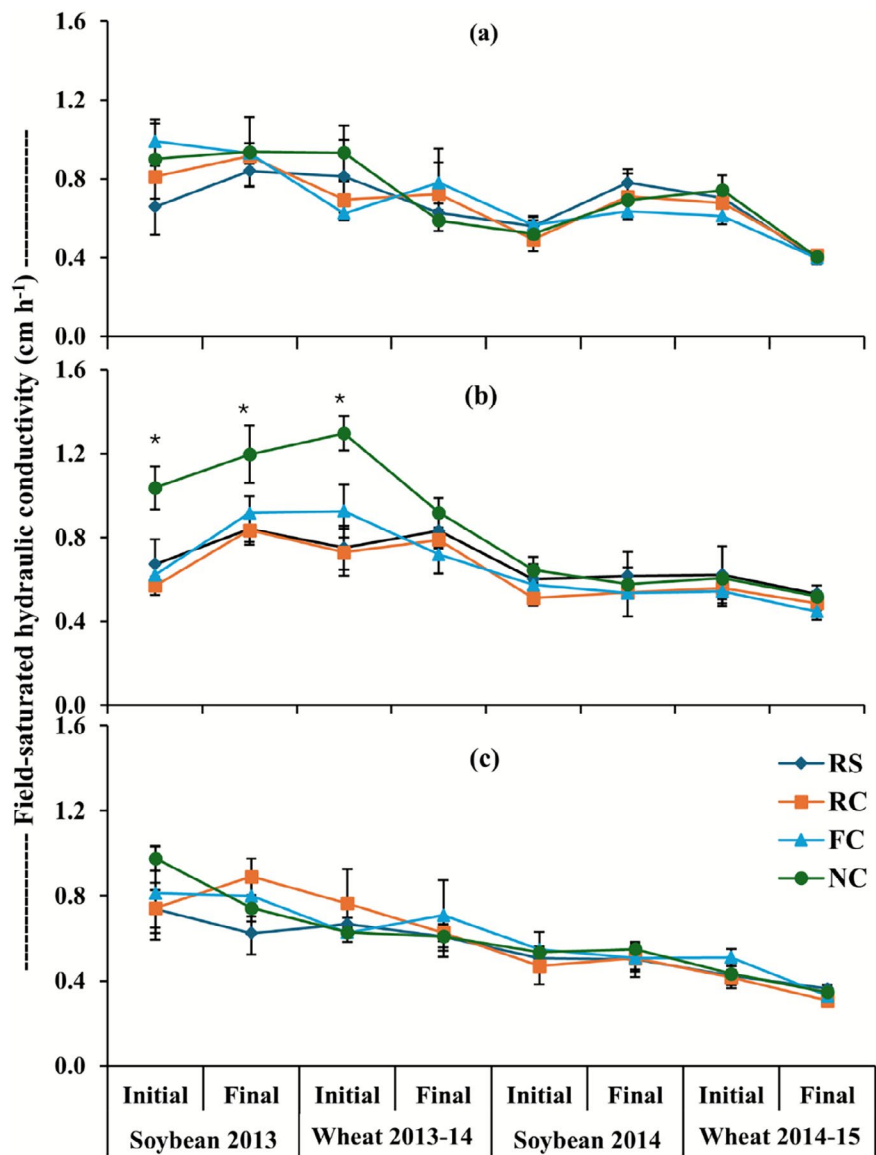


FIGURE 1 | Field saturated hydraulic conductivity (K_{fs} , cm h^{-1}) at (a) 0–15, (b) 15–30 and (c) 30–45 cm in soybean and wheat crop during 2013–14 and 2014–15 under chisel treatments: Residual (RS), repeated (RC), fresh (FC), and No chisel (NC) (points show treatment means; vertical bars are $\pm$ SEM. Significant treatment differences at $p=0.05$ within depth and sampling time are indicated as [*]).

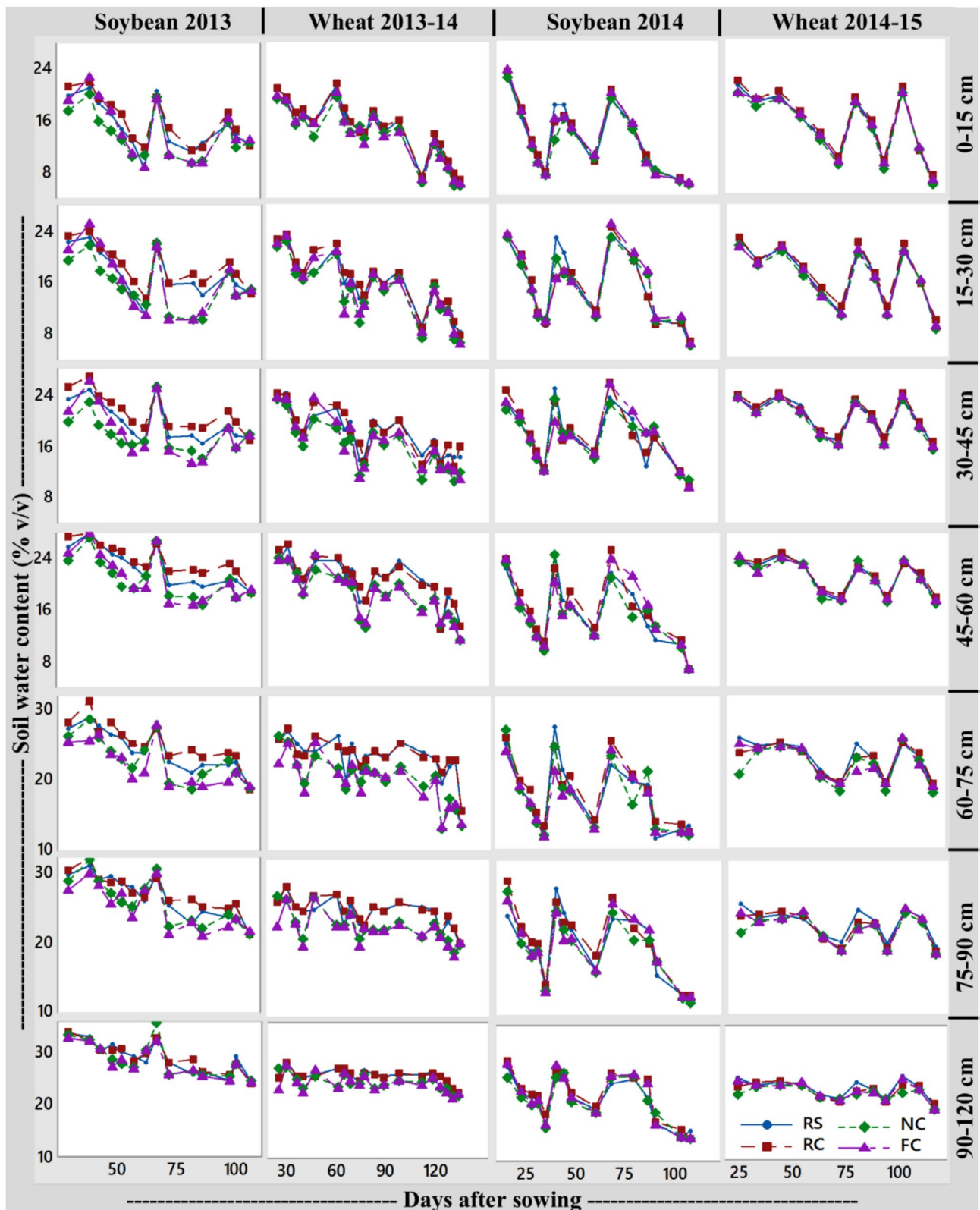


FIGURE 2 | Soil water content (SWC, % v/v) over the 2-year soybean-wheat rotation by depth, as affected by chiselling. FC, fresh chisel; NC, no chisel; RC, repeated chisel; RS, residual chisel.

were temporally consistent with the June–July storm loading that drove rapid reconsolidation in FC, helping explain why performance advantages did not persist. Crop water use increased by 11% (RC) and 8% (RS) compared to NC, resulting in broadly comparable water use efficiencies (WUEs) across treatments. Wheat

showed no significant difference in yields or water use, but WUEs were 18% and 5% higher in RC compared to FC and NC, respectively. In the second year, no differences in yield or WUE were observed for either crop, reinforcing the short-lived nature of chiselling benefits.

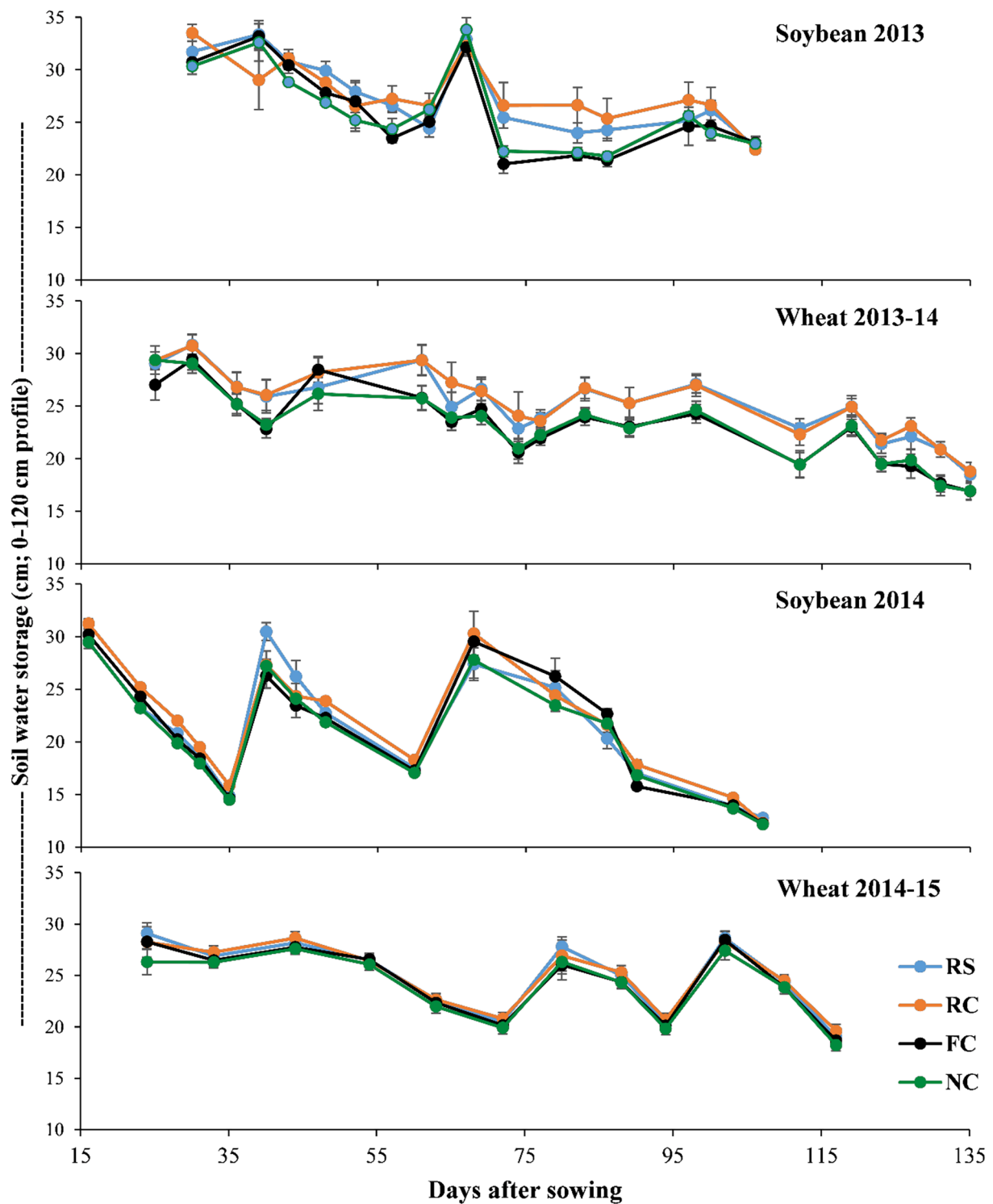


FIGURE 3 | Cumulative profile water content (0–120 cm), expressed as cm of water, over the experimental period (FC, fresh chisel; NC, no chisel; RC, repeated chisel; RS, residual chisel). Values are means $\pm$ SEM.

4 | Discussion

Sub-surface chiselling effectively reduced mechanical impedance, but benefits were most apparent early and diminished by the second year of the soybean-wheat rotation. Chiselling on 20 May 2013 was not followed by immediate storms; the first 14 days were dry. Instead, two concentrated storm clusters occurred 15–30 days (~145 cm) and 31–60 days

(~187 mm) after chiselling. This timing explains the observed pattern: early structural and hydraulic improvements in FC dissipated rapidly as intense storms acting on antecedent wet soil promoted slaking, small particles migration, settling and macropore collapse. By contrast, RS/RC, with older, partially re-consolidated loosening, retained water-storage advantages even as macropore continuity declined (Figure S1; Table S1).

TABLE 4 | Root length density (RLD, cm cm⁻³), root volume density (RVD, cm cm⁻³), root surface area density (RSAD, cm² cm⁻³) and average diameter (AD, mm) by depth (0–15, 15–30, 30–45 and 45–50 cm) under residual chisel (RS), repeated chisel (RC), fresh chisel (FC) and no chisel (NC) at harvest for soybean 2013, wheat 2013–14, soybean 2014, and wheat 2014–15.

Treatment	Length density				Volume density				Depth (cm)				Surface area				Average diameter				
	0-15	15-30	30-45	45-60	0-15	15-30	30-45	45-60	0-15	15-30	30-45	45-60	0-15	15-30	30-45	45-60	0-15	15-30	30-45	45-60	

Note: Values are mean (±SE).

TABLE 5 | Yield, seasonal water use, and water use efficiency (WUE) for (a) soybean (2013, 2014) and (b) wheat (2013–14, 2014–15) crops under fresh chisel (FC), residual chisel (RS), and repeated chisel (RC) and no-chisel (NC).

(a)						
Treatment	2013			2014		
	Yield	Water use	WUE	Yield	Water use	WUE
RS	1629 ^a	491 ^a	3.3 ^a	1270 ^a	568 ^a	2.2 ^a
RC	1783 ^a	504 ^a	3.5 ^a	1221 ^a	599 ^a	2.0 ^a
FC	1431 ^b	486 ^a	2.9 ^a	1093 ^a	580 ^a	1.9 ^a
NC	1351 ^b	452 ^b	3.0 ^a	1138 ^a	550 ^a	2.0 ^a

(b)						
Treatments	2013–14			2014–15		
	Yield	Water use	WUE	Yield	Water use	WUE
RS	4917 ^a	458 ^a	10.7 ^{ab}	3728 ^a	389 ^a	9.6 ^a
RC	4822 ^a	433 ^a	11.1 ^{ab}	4044 ^a	414 ^a	9.8 ^a
FC	4664 ^a	496 ^a	9.4 ^c	3551 ^a	387 ^a	9.2 ^a
NC	4758 ^a	451 ^a	10.5 ^b	3679 ^a	353 ^a	10.4 ^a

Note: Values are means; within each year and crop, numbers followed by different letters differ at $p=0.05$. Yield is in kg ha^{-1} , water use in mm, and WUE in $\text{kg ha}^{-1} \text{mm}^{-1}$.

Heavy rain can induce splash and downward movement of fine particles that block macro- and mesopores, reducing porosity and increasing soil BD, thereby accelerating re-consolidation. Although reduction in soil BD through deep ploughing or chiselling is widely reported, their persistence is limited under high-intensity rainfall. Reichert et al. (2009) observed a similar loss of tillage benefits in a Hapludalf soil following heavy rainfall. In the sandy and loamy sand soils of Punjab, deep tillage effects also declined after periods of rain or irrigation (Gajri et al. 1991). In another case, the benefits diminished after 115 cm of cumulative rainfall since the chiselling operation (Moriassi 2004). Recomposition of loosened soil layers by percolating water has been correlated to the cumulative rainfall received post-tillage (Busscher et al. 2002). Depending on rainfall intensity and duration relative to crop sowing, up to 60% chiselling benefits may be lost (Moriassi 2004). These findings underscore the temporary nature of chiselling, especially under no-till systems, as also noted by Botta et al. (2006), Silva et al. (2012) and Leão et al. (2014). In clayey soils under no-tillage, chiselling has also been shown to offer only short-term relief from compaction without leading to long-term improvements in soil structure (Nunes et al. 2015). Similarly, Nunes et al. (2019) found that chiselling effects in a Nitisol lasted only 12–24 months depending on depth. Traffic on loosened soil can further promote re-compaction, especially under high cumulative rainfall (Hamza and Anderson 2005; Busscher et al. 2002; Silva et al. 2012).

In the present study, PR reduced by 26%–30% immediately after chiselling on 28 May 2011 in the same field (Personal communication; not part of this study), but such benefits were largely lost in freshly chiselled plots due to subsequent heavy rainfall. In contrast, early chiselling (RS and RC) likely allowed the soil to approach a new structural equilibrium before the 2013 storms,

conferring greater resistance to reconsolidation. These observations reinforce that the timing and frequency of chiselling must align with regional rainfall patterns, especially in semi-arid systems where rainfall is often intense, erratic and poorly distributed.

K_{fs} decreased with depth, as expected due to increased compaction and reduced pore connectivity in deeper soil layers. Yet in the first soybean season, NC showed greater K_{fs} at 15–30 cm, contrary to many studies reporting improvements in K_{fs} after chiselling (Sauer et al. 1990; Moreno et al. 1997; Miller et al. 1998; Cameira et al. 2003). This pattern, seen in both IR and K_{fs} , indicates a dual-porosity situation. In NC, flow at near-saturation is controlled by a few continuous macropores, which give high IR and K_{fs} , while PR reflects the strength of the surrounding matrix between those pores. Thus, high IR/ K_{fs} (macropore-controlled) can occur alongside high PR (matrix-controlled). In FC, chiselling lowers PR by loosening the matrix and increasing pore numbers, but it disrupts vertical continuity of the pre-existing macropores. The network becomes more fragmented—good for storage (greater SWC in RS/RC on several dates), but poorer for fast through-flow (lesser or short-lived IR/ K_{fs}). High-intensity rainfall soon after chiselling likely settled fine particles into loosened pores and promoted slaking/settling, especially in FC and RC, further reducing IR and K_{fs} ; by contrast, NC retained structurally stable macropore networks that favoured rapid infiltration and deep percolation, albeit at the cost of lesser SWC. Similar rainfall driven declines in K_{fs} to pre-tillage levels have been documented (Moriassi 2004), whereas transient increases in profile water content after deep tillage are well reported (Boydas and Turgut 2007; Jawadi et al. 2009; Mari et al. 2011).

Chiselling moderately increased SWC across depths, particularly in RC and RS, with benefits most visible in year 1. However,

gains were not consistently maintained in year 2, consistent with weather-driven re-consolidation (Boydas and Turgut 2007; Jawadi et al. 2009; Mari et al. 2011). The marginally greater profile water content (0–120 cm) in RC (~20.6 cm) indicates some lasting storage benefit, but not enough to affect second-year crop performance.

Chiselling also improved root proliferation, particularly in the 15–30 cm compacted zone during year 1. Both RS and RC increased RLD and RSAD, particularly in wheat. This is supported by findings from Moraes et al. (2020), who observed that in compacted soils, mechanical impedance was the primary constraint to soybean root growth, whereas in loosened soils, matric potential (water stress) became the dominant limiting factor. The compact 15–30 cm layer reduced root length, volume and roots also became thinner in wheat. This confirms that reduced sub-surface strength supports better root architecture, consistent with findings from Rosolem et al. (2002) and Mondal et al. (2019). Shallow proliferation of roots (0–15 cm) in NC/FC plots likely reflects sub-surface restriction. Soybean, being a shallow-rooted crop, showed less response, whereas wheat, with its deeper root system, was more sensitive to changes in sub-soil conditions. Indeed, field studies have shown that 50%–60% of wheat roots may remain confined within the top 15 cm under compaction, resulting in up to 23% yield loss (Patel et al. 2020). These results are in line with recent studies highlighting species-specific rooting responses to physical barriers (de Moraes et al. 2016).

Despite improvements in root growth and water content in year 1, yield and WUE gains were significant only for soybean, particularly under RS and RC. Although wheat has a deeper root system and our data show greater root growth at 15–30 cm in RS/RC, this did not translate into higher wheat yield. Two points likely explain the contrast. First, the heavy mid-June to early-July storms rapidly reduced the hydraulic advantages of FC, so by the wheat season many structural differences had already weakened—the extra root growth in wheat therefore operated under smaller functional gains. Second, during the wheat crop, water stress was not consistently limiting under our conditions (profile water and management were similar among treatments), so additional rooting did not improve grain yield. By contrast, soybean, which relies more on the upper profile, benefited early from reduced impedance near 15–30 cm, improving water access and growth during critical stages. In short, wheat roots explored deeper soil, but the combination of attenuated structural differences and limited water constraint meant that extra roots did not raise yield, whereas soybean's shallower system captured a short-lived but meaningful advantage. Similar outcomes have been reported in other semi-arid and subtropical cropping systems, where tillage-induced gains are short-lived unless coupled with moisture-conserving practices or residue retention (Alvarez and Steinbach 2009; Leão et al. 2014).

Overall, the study highlights that the benefits of sub-soil chiselling on soil structural properties, water retention and root development are highly dependent on timing, rainfall conditions and the duration since tillage. While residual and repeated chiselling improved sub-surface conditions and early crop performance, these effects were short-lived and largely diminished by year 2. These findings emphasize the need for site-specific and rainfall-informed chiselling strategies to enhance soil structural

resilience and sustainable crop productivity, particularly in semi-arid agroecosystems prone to rapid reconsolidation.

Soil PR could not be recorded at soybean harvest (2013) and at early wheat (2013–14) due to penetrometer malfunction. Because PR is our primary indicator of mechanical impedance, this gap increases uncertainty about the timing and persistence of sub-surface strength differences during that interval. We therefore triangulated this period with co-measured proxies. Before the gap, PR already differed among treatments (RS and RC < NC; FC intermediate). During the gap, hydraulic response (IR , K_{fs}) tended to be greater in NC, whereas RS and RC showed greater SWC on several dates—evidence of functional loosening and water-storage advantages even where macropore continuity may have been transient. Root metrics (e.g., higher RLD at 15–30 cm in RS/RC) are consistent with these observations. Integrative outcomes (yield, WUE) improved only in year 1 in soybean under RS/RC and attenuated thereafter, mirroring rainfall-driven re-consolidation. The missing PR limits our quantification of the magnitude and rate of impedance rebound in late soybean 2013 and early wheat 2013–14 but does not alter the central inference that chiselling effects are brief under heavy rainfall and more persistent under residual/repeated chiselling than under FC.

5 | Conclusions

This study shows that the effects of sub-surface chiselling on soil structure, water retention and root development were short-lived and highly sensitive to rainfall timing and intensity. Freshly chiselled plots lost most benefits after intense storms that occurred weeks after tillage, whereas residual and repeated chiselling offered more sustained improvements in water storage despite reduced macropore continuity. Together with the daily rainfall analysis, these results suggest that timing relative to high-intensity rainfall events largely determines how long chiselling benefits last.

Our two-season dataset does not justify prescribing a fixed interval. Very frequent chiselling is unlikely to be efficient (benefits attenuated within the first season under heavy storms), yet some RS/RC signals persisted into the following year without consistent yield gains. A monitor-and-trigger approach is therefore more defensible than a calendar rule—re-chisel only when soil strength (PR/BD) and/or hydraulic function (IR/K_{fs}) cross site-specific thresholds. For sandy loam soil under similar rainfall regimes, reassessment after ~3–4 years is reasonable as a working hypothesis because it avoids annual disturbance yet checks for renewed impedance. However, it requires validation in longer trials. Integrating chiselling with conservation practices (e.g., residue retention, controlled traffic) may help extend benefits while limiting disturbance.

Author Contributions

Mukhtar Ahmad: methodology, data curation, validation, formal analysis, investigation. **Pramila Aggarwal:** methodology, resources, writing – review and editing, software, validation. **R. N. Garg:** methodology, resources, validation. **Surajit Mondal:** writing – review and editing. **Sunayan Saha:** writing – review and editing. **Debasish Roy:** writing – review and editing, validation. **Ravender Singh:** supervision,

investigation, validation. **Debashis Chakraborty**: conceptualization, investigation, supervision, funding acquisition, project administration, resources, writing – review and editing.

Acknowledgements

Authors thank the Director, ICAR-IARI, New Delhi for providing the necessary facilities for the study. D.C. acknowledges ICAR-IARI and the ICAR-Extramural Fund. M.A. was supported by a Fellowship from ICAR-IARI. During the preparation of this work the author(s) used ChatGPT only in the writing process to improve the readability and language of the manuscript. After using this tool/service, the author(s) reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding

This research was financially supported by the Institute Core Project and the Extramural fund from the Indian Council of Agricultural Research.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supporting information.